Radial Consolidation: Constraining Post-Formation Kernel Drift in Spiking Neural Networks

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***Abstract*—Continued local plasticity can preserve linearly readable information while degrading decisions made directly from spike timing. We investigate this discrepancy in two-layer spiking networks with label-free hidden-layer STDP and a supervised temporal classifier. On MNIST, late hidden kernels retain high linear Readout accuracy but support worse native decisions even after their weights stop changing. A four-condition kernel/homeostasis study associates this cost with changes in direction, spatial DC/contrast balance, and spectrum, without identifying rotation as the sole cause. A longer trajectory also shows directional drift after aggregate geometry has nearly stabilized. We then test reference-direction constraints on post-formation updates. Radial-only updates preserve the component parallel to each receiver's reference kernel while suppressing its orthogonal component. At epoch 20, the MNIST validation difference from hard freezing is $+0.057$ percentage points (95% paired CI $[-0.043, 0.157]$, five seeds), although correct-class evidence remains later. On N-MNIST, seven seeds yield a validation difference of -0.007 points and a test difference of $+0.111$ points, with improved temporal separation. Constraints imposed from initialization or on the classifier do not share this benefit. These results support stage- and layer-specific consolidation, not a universal solution to local learning. Because normalization can cancel radial changes, the present proof of concept establishes decision preservation with an enabled update pathway, not useful lifelong adaptation.**

***Index Terms*—spiking neural networks, local plasticity, STDP, temporal coding, representational drift, consolidation**

I. Introduction

Learning a representation and preserving its usefulness to a downstream circuit are different objectives. This distinction is especially important for spiking neural networks (SNNs) whose prediction is the first spike of a class-specific output group. An external classifier can reweight a population representation and recover class information without inheriting the original circuit's thresholds, competition, or temporal integration. High external-readout accuracy therefore does not establish that the same representation remains usable by the network's native decision mechanism. Work on compensating representational drift shows that stable communication depends on how a downstream circuit tracks the changing code, not simply on whether that code changes [1].

Local spike-timing-dependent plasticity (STDP) provides a practical route to learning without backpropagating a task gradient through hidden layers [2]. Synaptic scaling and coordinated plasticity mechanisms already offer ways to stabilize activity and support memory [3], [4]. However, a hidden local rule need not know whether its next update will preserve the downstream temporal margin. Its synaptic activity statistics can remain sufficient to drive updates after a useful representation has formed. This is a mismatch between the information available to the tested rule and the downstream property to be preserved; it is not a claim that all local learning rules must update forever or that gradient-based systems are automatically stable.

We study the resulting *post-formation kernel drift* in a controlled two-layer system. The hidden cortex, denoted $A-1$, learns without labels. The output cortex, A, learns class decisions from relative spike timing. We use hard freezing of the hidden cortex after a short formation period as a stabilization reference. This intervention disables both its kernel learning and its homeostatic working-point adaptation. The scientific question is whether that complete shutdown is necessary.

Our central question is: *Can harmful changes orthogonal to formed kernel directions be restricted without completely disabling the local update pathway?* We decompose each update relative to a stored receiver-specific kernel and attenuate its orthogonal component. The radial-only endpoint retains the parallel component. Parameter consolidation and gradient projection provide relevant precedents [5], [6], [7], but here the protected object is a formed receiver-specific kernel direction, without a hidden task-gradient importance estimate. This is a deliberately simple intervention: its value lies in testing a mechanistic hypothesis and defining the limits of a consolidation strategy, rather than in the novelty of vector projection itself.

The evidence supports three contributions. First, we distinguish readable information from native temporal usability using kernel-state controls, activity diagnostics, and separate correct/wrong timing measurements. Second, we show that receiver directions can continue changing when coarse spectral and DC statistics have nearly plateaued. Third, we evaluate direction-constrained consolidation on MNIST and native N-MNIST events, including independent seeds, formation-time and layer-placement controls, and explicit normalization and hardware caveats. The contribution is a bounded proof of concept for post-formation stabilization, not a demonstrated continual-learning algorithm or a benchmark state-of-the-art claim.

II. Related Work and Scope

STDP classifiers combine synaptic learning with competition and activity regulation. Diehl and Cook [2] demonstrated

unsupervised digit learning with STDP, adaptive thresholds, and lateral inhibition. For supervised spike-native classification, neuronal competition groups assign multiple competing neurons to each class and regulate their participation [8]. These studies motivate our ingredients, but neither a healthy firing distribution nor successful feature extraction alone guarantees preservation of a downstream temporal margin during continued hidden plasticity.

A. Activity stabilization and native decision preservation

Homeostatic regulation addresses an important but different objective. Synaptic scaling can stabilize activity while interacting with Hebbian changes [3]. Zenke, Agnes, and Gerstner [4] combined Hebbian learning with rapid heterosynaptic compensation, slower homeostatic changes, and consolidation to support assembly formation and memory recall in a recurrent spiking network. Such results rule out framing our problem as an absence of stabilization mechanisms for local learning. The question is instead which functional property a particular combination of mechanisms preserves.

In our artificial circuit, threshold and gain regulation control the operating point; they do not directly optimize the hidden representation for downstream classification. Preserving firing activity therefore does not by itself establish preservation of correct-versus-wrong spike ordering. We test homeostasis separately instead of assuming that it is always either the source of deterioration or a successful remedy, and assess temporal evidence alongside accuracy and activity. This is a diagnostic distinction, not a claim that our intervention reproduces biological synaptic scaling or replaces the mechanisms in those models.

B. Representational drift and compensating decoders

Representational drift is not intrinsically harmful. Rule and O'Leary [1] showed how Hebbian and homeostatic mechanisms can exploit population-code redundancy to track gradual representational changes without an external error signal. Their work provides a constructive example of compensation rather than a reason to regard changing neuron tuning as inevitable functional failure.

Our concern is narrower: continued drift under a particular local rule and native temporal decoder, without a guarantee of such compensation. We distinguish a linear Readout fitted to hidden features from decisions made by the trained spike-timing circuit. Agreement or disagreement between these decoders tests different aspects of the representation; high Readout accuracy is not evidence that native timing has been preserved. Fixed-kernel schedules and freshly initialized classifiers further examine whether a late hidden state remains costly after ongoing kernel motion is removed. These controls do not test the self-healing mechanism itself. Likewise, matched-neuron directional change is not necessarily a rotation of the population's principal subspace; we use receiver directions as an operational diagnostic rather than a substitute for principal-subspace measurements.

C. Consolidation, projection, and information requirements

Continual-learning methods protect previously useful parameters or update subspaces. Elastic weight consolidation penalizes parameter displacement according to importance estimated with a diagonal Fisher approximation [5]. Synaptic Intelligence accumulates each parameter's contribution to changes in the task loss along the learning trajectory [6]. Although that accumulation is parameter-wise, its dependence on the task gradient distinguishes it from our label-free hidden update. Gradient Projection Memory uses SVD of network activations to construct retained bases and projects new task gradients orthogonally to the protected subspaces [7].

Our operation instead suppresses directions orthogonal to a receiver's own reference *weight* vector. In particular, "orthogonal" describes different reference objects: GPM protects activation-derived subspaces, whereas radial-only retains the component along a stored kernel direction. Our rule requires neither hidden task gradients nor a training-time SVD, but it stores a reference kernel and performs a receiver-wise reduction. It is thus neuron-local in its information dependency, not strictly autonomous at each synapse. The supervised output layer still receives class information, so we do not characterize the entire system as label-free.

The evaluation also differs: we test post-formation stabilization during repeated stationary classification, not retention across a sequence of new tasks. These methods establish context, not matched performance baselines. We neither claim to solve catastrophic forgetting nor compare accuracy directly against unmatched gradient-trained models. Our present contribution is to diagnose loss of native temporal usability and test a restricted consolidation intervention; useful adaptation after a distribution shift remains unverified.

III. Model, Intervention, and Measurements

We distinguish the native decision measure and baseline learning rules below from the proposed direction constraint in Algorithm 1. The subsequent normalization analysis states a limitation of the intervention rather than an additional learning rule.

A. Native temporal classifier

Inputs drive a convolutional hidden cortex followed by a class-group output cortex, without direct input-to-output skip connections. Neurons integrate weighted input, emit at most one spike per sample, and participate in lateral competition. Hidden outputs are max-pooled before entering the classifier. For a neuron firing at time t_i within a simulation horizon T, its normalized earliness is

$$e_i = \begin{cases} (T - t_i)/T, & \text{if it fires,} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Let $\mathcal{G}_c$ be the output group for class c. Its score is $E_c = \max_{i \in \mathcal{G}_c} e_i$. For label y,

$$E_{\text{correct}} = E_y, \qquad E_{\text{wrong}} = \max_{c \neq y} E_c, \qquad E_{\text{gap}} = E_{\text{correct}} - E_{\text{wrong}}. \tag{2}$$

Prediction uses the native maximum group score; an all-silent output is separately counted as no decision. Larger E means *earlier* evidence, not longer latency. A negative change in E_{correct} means that correct evidence arrives later, whereas a negative change in E_{wrong} means that competing evidence arrives later.

Balanced pre- and postsynaptic traces have the form $\bar{s}(t) = \gamma\bar{s}(t-1)+(1-\gamma)s(t)$. The label-free hidden update combines potentiation and depression,

$$u_{ij} = \eta_+ \sum_t \bar{s}_i(t)s_j(t) - \eta_- \sum_t s_i(t)\bar{s}_j(t). \tag{3}$$

At the classifier, signed potentiation is modulated by a temporal-margin teaching signal. For class c and margin m, define

$$h_c = \begin{cases} \max(0, m - E_{\text{gap}}), & c = y, \\ \min(0, E_y - E_c - m), & c \neq y. \end{cases} \tag{4}$$

The correct class receives a positive signal until its lead reaches m; each competing class receives a negative signal if it is not sufficiently behind. Within a group, $\rho_j = \exp(e_j/\tau)/\sum_{k\in\mathcal{G}_c}\exp(e_k/\tau)$ allocates the signal, with separate positive/negative gains. The output update is the potentiation term of (3) multiplied by the resulting signed modulation. A decision pass precedes replay with fixed weights; updates accumulate and are applied after the replay. Labels are not propagated into $A-1$.

B. Reference-direction constraint

We call the radial-only intervention *radial consolidation*: a post-formation constraint on existing STDP updates, evaluated as an alternative to fully freezing the hidden cortex rather than a replacement for formation-stage learning. For receiving neuron j, let w_j contain its incoming kernel coefficients on learnable connections, and let r_j denote this kernel at the formation boundary. Algorithm 1 defines the complete constrained update, applied separately to each receiver; receiver subscripts are omitted there. LocalSTDP denotes the update in (3), normalize uses the L_2 norm clamped below at 10^{-12}, and Regulate denotes the existing kernel regulation $\mathcal{R}$. Thus a zero reference contributes no parallel component. The normalized reference q_j remains fixed throughout subsequent training.

The projection term retains the update parallel to q_j, while α scales the orthogonal component (Fig. 1). We test $\alpha \in \{1, 0.5, 0.25, 0.1, 0\}$: $\alpha = 1$ is unrestricted and $\alpha = 0$ is radial-only. The tested direction is a *receiver's incoming receptive field*, not a separately learned outgoing axonal direction. Hidden homeostasis and output-classifier learning remain enabled according to the training protocol.

For P learnable coefficients, the intervention requires $O(P)$ reference storage and $O(P)$ arithmetic per update, including one dot-product reduction per receiver. This states the algorithmic overhead; no hardware energy or latency benefit is measured.

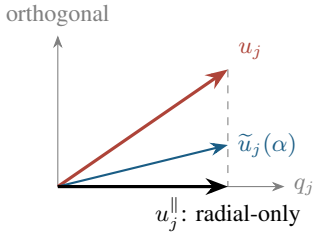


Fig. 1. Reference-direction consolidation. The local update u_j is constrained toward the fixed reference q_j; $\alpha = 0$ retains only its parallel component. This conceptual sketch is not an empirical kernel projection.

Algorithm 1 Reference-constrained hidden update

At formation boundary:
$q \leftarrow \text{normalize}(w)$ // Store once

For each subsequent update:
$u \leftarrow \text{LocalSTDP}(\text{spike traces})$
$u \leftarrow \alpha u + (1-\alpha)q(q^\top u)$
$w \leftarrow \text{Regulate}(w + u)$

We compare this intervention with *hard freeze*, which disables both hidden kernel learning and hidden homeostasis while the output cortex continues learning. Radial-only instead continues to compute and project hidden local updates while allowing hidden homeostatic adaptation. We also study kernel-only freezing with homeostasis continuing. These are different interventions; the radial-only versus hard-freeze contrast is not a pure comparison of weight-update directions.

An intervention at @k is applied *before training epoch* k. Thus @2 follows one completed formation epoch, while @10 follows nine. A final epoch-20 measurement follows twenty completed epochs. We distinguish these intervention boundaries from checkpoints saved after a specified number of completed epochs.

C. Normalization and the meaning of retained plasticity

Both cortices use per-receiver L_1 normalization after updates. This creates a crucial limit on the radial-only interpretation. If $w_j = c_j r_j$ with $c_j > 0$, the radial increment merely changes c_j. Ideal fixed-target normalization satisfies

$$\mathcal{N}_L(c_j r_j) = L\frac{r_j}{\|r_j\|_1}, \qquad c_j > 0. \tag{5}$$

Consequently, a nonzero proposed or projected update can leave the normalized kernel unchanged. Weight floors, masked connections, and other regulation can produce small residual differences, but an enabled STDP pathway does not establish useful kernel adaptation.

We therefore distinguish the number of update steps, the number with a nonzero raw update, the projected/raw update norm ratio, and post-regulation drift. A step with a nonzero raw update need not produce a nonzero realized weight change. We interpret radial-only as a consolidation endpoint; raw-update activity alone is not evidence of lifelong learning.

D. Geometric and functional diagnostics

For matrix W formed from flattened receiver kernels, let s_ℓ be its singular values. We report the participation ratio

$$R_{\mathrm{PR}} = \frac{(\sum_\ell s_\ell^2)^2}{\sum_\ell s_\ell^4}. \tag{6}$$

This is a spectrum summary, not an entropy-based effective rank. For MNIST kernels, DC is the spatial mean *within each of the six input channels*. If W_{DC} repeats those channel means spatially, DC energy is $\|W_{\mathrm{DC}}\|_F^2/\|W\|_F^2$ and contrast energy is its complement. Signed mass is $\sum W/\sum |W|$, not the fraction of positive entries. We additionally measure matched-receiver cosine, relative weight displacement, and the fraction of displacement orthogonal to the reference. Cross-arm kernel comparisons and within-run temporal trajectories are identified separately.

Functional assessment includes native validation/test accuracy, no-decision rate, sample-level margins, per-class correct/wrong evidence, hidden and output silence, and activity concentration. Positive mean margins alone are insufficient: an all-silent circuit can misleadingly have no negative margins. Linear Readout models fitted to hidden earliness features are diagnostic instruments, never the native decision rule. They are available for MNIST but were disabled in the N-MNIST intervention batch.

IV. Experimental Protocol

We use MNIST [9] as a controlled temporal-coding testbed and N-MNIST [10] for native event input. Each uses a seed-specific 90/10 split of the original training set, preserving its test set. Mini-batches contain 200 samples. Unless stated otherwise, comparisons use final epoch 20, not the best test checkpoint. Test results are secondary outcomes; reported sweeps are not used to claim a universally optimal α.

Table I gives the principal settings. MNIST uses six ON/OFF difference-of-Gaussian channels, with filter sizes 3, 7, and 13, followed by rank-order encoding over 40 steps. N-MNIST uses both polarities at native 34×34 resolution, binary occupancy in 20 temporal bins, and the complete recording rather than only its first saccade. Hidden STDP has $\eta_+ = 10^{-6}$ and $\eta_-/\eta_+ = 0.7$. There is no backpropagated hidden task gradient.

Threshold homeostasis remains adaptive outside freeze interventions. Amplifier adaptation is enabled for MNIST but disabled for N-MNIST, whose amplification is fixed at 10. Thus "homeostasis continues" in the latter does not mean that both threshold and amplifier are adapting. MNIST Readout models use a linear classifier on earliness features, fitted with L-BFGS, weight decay 10^{-4}, and at most 200 iterations.

The principal MNIST sweep contains five paired seeds (34160–34164); an independent layer-placement batch uses five further seeds (34165–34169). A fresh initialization-boundary comparison uses seven seeds (34170–34176). The kernel/homeostasis study uses four seeds (34200–34203), and the N-MNIST intervention uses seven (34220–34226). A separate long trajectory uses four seeds (34100–34103).

TABLE I
Principal protocol settings. The output classifier has ten class groups. Hidden pooling uses 2×2 max pooling; output pooling is global.

Setting	MNIST	N-MNIST
Input channels / time steps	6 / 40	2 / 20
Hidden channels	300	330
Hidden kernel / stride	10×10 / 2	9×9 / 2
Output neurons per class	20	45
Trace decay γ	0.66	0.97
Lateral inhibition factor	3	30
Temporal margin m	0.30	0.35
Positive / negative teaching gains	40 / 120	32 / 100
Group softmax temperature τ	0.25	0.25
Threshold bounds	$[0.1, 10]$	$[1, 100]$
Threshold log-step	0.00125	0.00125
Hidden amplifier log-step	$6.25 \cdot 10^{-6}$	0
Output amplifier log-step	$1.25 \cdot 10^{-6}$	0
Per-receiver normalization	L_1	L_1

The receiver-mask comparison uses five seeds. Each reported comparison includes all stated seeds at its specified evaluation endpoint.

For each comparison we compute seed-wise differences and a two-sided Student-t 95% confidence interval (CI). Accuracy differences are in percentage points (pp); timing is dimensionless normalized earliness. Seed pairing does not imply identical hardware execution: GPU models differ between some arms. For radial-only versus hard freeze, same-GPU counts are 2/5 in the main MNIST batch, 0/5 in its replication, and 2/7 in N-MNIST. The corresponding MNIST radial-only versus unrestricted counts are 1/5 and 4/5. In the kernel/homeostasis study, early/late pairs share GPU models in 3/4 continuing-homeostasis and 1/4 frozen-homeostasis pairs. These CIs quantify observed paired-seed variation; they do not remove confounding from differences in hardware or software versions between runs. Small secondary differences should be interpreted accordingly. The multiple exploratory contrasts are not family-wise corrected.

V. Results

A. Stable coarse geometry does not imply stable directions

Figure 2 visualizes matched-neuron directions and their full-dimensional angular change throughout training. The long MNIST trajectory reaches epoch 100 with native test accuracy 91.655% (95% CI $[90.059, 93.251]$), while the hidden linear Readout reaches 98.485%. These are different decoders of the representation, not interchangeable estimates of model accuracy. Table II separates aggregate kernel statistics at each checkpoint (A) from directional changes between checkpoints (B). From epoch 70 to 90, participation ratio, global DC energy fraction, mean receiver DC fraction, and signed mass change by only -1.225%, $+1.084\%$, $+0.105\%$, and $+0.176\%$, respectively. Yet the same interval has mean direction cosine 0.722, mean angle 36.0°, and orthogonal fraction 90.9%. Thus nearly stable aggregate statistics can conceal substantial

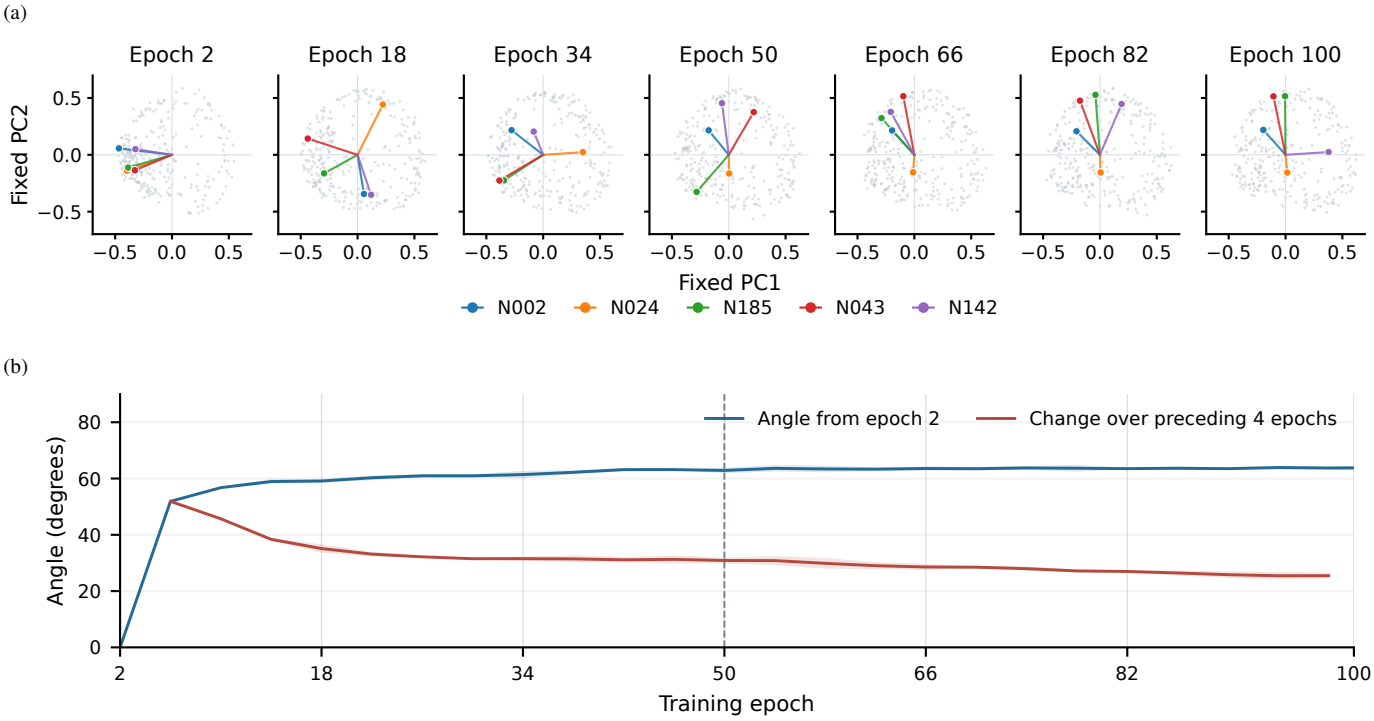


Fig. 2. Continued kernel-direction drift under local plasticity. (a) Five neurons uniformly sampled without replacement from the 300-neuron hidden layer of one MNIST run, tracked at epochs 2, 18, 34, 50, 66, 82, and 100. No rotation-angle, quadrant, or performance filter was used. Each incoming kernel is normalized to unit L_2 norm and projected into a single centered PCA basis fitted to all saved checkpoints of this run. The two displayed components explain 30.3% of centered variance. Neuron identities, colors, axes, and scale remain fixed; gray points show all 300 neurons. Spokes originate at the common PCA mean, not at the physical zero kernel, so their projected angles are not full-dimensional kernel angles. (b) Angles in the original 600-dimensional kernel space, relative to epoch 2 (blue) and to the preceding saved checkpoint four epochs earlier (red). Angles are averaged over all 300 neurons within each seed; curves and shaded bands show the mean and Student-t 95% confidence interval across four seeds. The terminal two-epoch interval is excluded from the red curve. The dashed line marks model-state continuation at epoch 50 without restored RNG or data-loader state; runs may use different GPU models. Nonzero local angular change can persist while distance from the early reference plateaus. This visualization describes drift and does not alone establish its causal effect on native decisions.

changes in individual kernel directions. Direction angles use the full 600-dimensional kernels; unchanged directions have cosine 1 and angle $0°$.

This supports persistent directional change over the observed interval, not indefinitely increasing rank or a theorem of nonconvergence. A separate terminal check from epoch 90 to 100 gives matched-receiver cosine 0.7747 and mean orthogonal fraction 92.32%, with nonzero displacement in all 300 receivers. Over this terminal interval, the change in mean temporal gap is -0.00234 with CI $[-0.01106, 0.00638]$. Thus late drift is not accompanied by a clearly established further mean-gap decline over that interval. Nor is it simply a dead network: epoch-100 no-decision is 1.175%, all 40 seed–class mean gaps are positive, and hidden top-decile activity share is 0.1429. The trajectory resumes model state at epoch 50 without restoring the original random-number-generator/loader state and may change GPU; it is not a bitwise uninterrupted 100-epoch training run.

B. A fixed late kernel still carries a native decision cost

The four-condition study crosses an early/late hidden-kernel freezing boundary (@2 or @10) with continued/frozen hidden homeostasis. All conditions train the output cortex through epoch 20. This is a matched-design schedule comparison, not a transplant of an identical kernel into independently randomized homeostatic states. Exact parameter checks verify that every hidden kernel is unchanged throughout the common epoch-10-to-20 evaluation interval.

With continuing homeostasis, the late-minus-early validation difference is $-1.329\,\text{pp}$ (CI $[-1.879, -0.779]$); with frozen homeostasis it is $-1.379\,\text{pp}$ ($[-2.078, -0.680]$). The corresponding test differences are $-1.248\,\text{pp}$ and $-1.220\,\text{pp}$. Hidden Readout models remain between 98.51% and 98.76% (Fig. 3). The persistent native cost therefore cannot be explained solely by the classifier currently chasing a moving *kernel*. Other hidden-state dynamics and training history are not thereby excluded.

A separate checkpoint experiment addresses inherited classifier history more directly. Hidden states saved after two or ten *completed* epochs are fully frozen, and a same-seed freshly initialized output cortex is trained for ten epochs (four seeds, 34000–34003). Early minus late hidden state yields $+0.62\,\text{pp}$ validation (CI $[0.31, 0.94]$) and $+0.51\,\text{pp}$ test ($[0.41, 0.61]$). Reinitializing rather than retaining the output classifier on the late state changes validation by only $+0.03\,\text{pp}$ ($[-0.27, 0.33]$). This finite-budget control supports a persistent hidden-state

TABLE II

NEARLY STABLE AGGREGATE STATISTICS COEXIST WITH CONTINUED DIRECTIONAL DRIFT (FOUR SEEDS). (A) MEAN KERNEL STATISTICS AT EACH CHECKPOINT. (B) DIRECTIONAL CHANGES OVER THE LABELED 20-EPOCH INTERVALS, AVERAGED OVER 300 MATCHED RECEIVERS PER SEED, THEN ACROSS SEEDS. ORTHOGONAL FRACTION IS ORTHOGONAL DISPLACEMENT NORM DIVIDED BY TOTAL DISPLACEMENT NORM, RELATIVE TO THE INTERVAL'S STARTING DIRECTION, EXPRESSED AS A PERCENTAGE. IT IS NOT AN ENERGY OR NEURON-COUNT FRACTION. THE 50–70 INTERVAL CROSSES A MODEL-STATE RESTART (SEE TEXT).

(A) Aggregate kernel statistics at each checkpoint

Statistic	Epoch 10	Epoch 30	Epoch 50	Epoch 70	Epoch 90
Participation ratio	4.90975	5.54409	5.78613	6.02215	5.94795
Global DC energy fraction	0.40941	0.37834	0.36742	0.36042	0.36432
Mean receiver DC fraction	0.48810	0.45970	0.44127	0.43116	0.43160
Signed mass	0.99768	0.99395	0.99248	0.99244	0.99419

(B) Directional change over consecutive 20-epoch intervals

Statistic	Baseline	10 → 30	30 → 50	50 → 70	70 → 90
Mean direction cosine	-	0.536	0.620	0.671	0.722
Mean direction angle (0-180°)	-	55.6°	47.5°	42.1°	36.0°
Orthogonal fraction (0-100%)	-	87.2%	88.8%	89.9%	90.9%

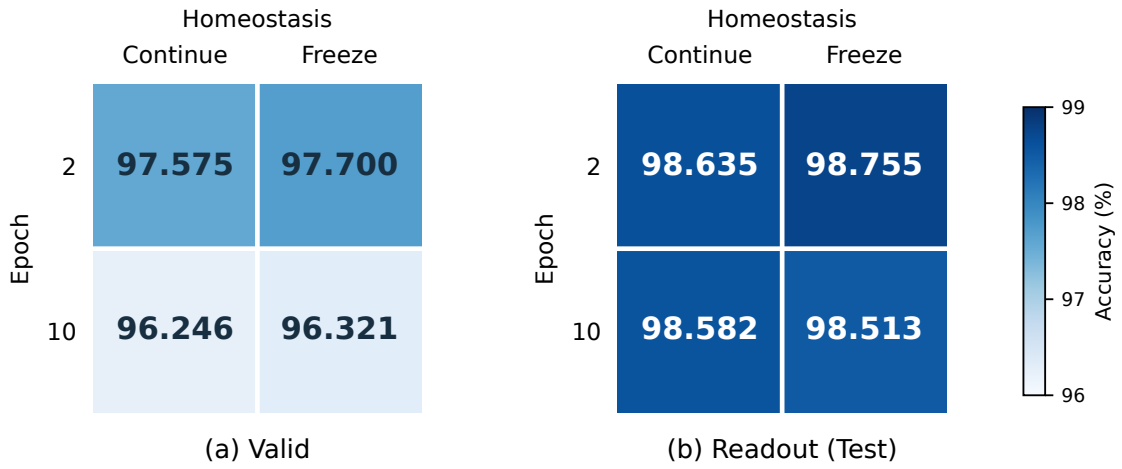


Fig. 3. Kernel boundary and homeostasis (four paired seeds). Rows indicate the epoch before which hidden-kernel learning is frozen; columns indicate continued or frozen hidden homeostasis. All conditions are evaluated at epoch 20. Cells show mean accuracy (%). Left: native validation (Valid). Right: hidden linear Readout test accuracy (Readout). Both panels share a truncated 96–99% scale. The panels use different data splits, not a paired validation-to-test comparison. This matched schedule comparison does not isolate pure rotation, and GPU models are not fully matched.

cost, rather than an explanation based solely on obsolete classifier weights. It does not establish an asymptotic upper bound for a fresh classifier, and the two saved hidden states differ in homeostasis as well as kernel structure.

The final timing deficit is primarily delayed correct evidence. Under continuing homeostasis, late-minus-early $\Delta E_{\text{correct}} = -0.02040$ (CI $[-0.02390, -0.01690]$), whereas $\Delta E_{\text{wrong}} = +0.00006$ ($[-0.00292, 0.00305]$). Under frozen homeostasis, $\Delta E_{\text{correct}} = -0.04682$ and $\Delta E_{\text{wrong}} = -0.00122$. Earlier wrong-class evidence is therefore not a sufficient description of the final deficit; its relevance can differ across training stages.

Homeostasis also is not an unconditionally beneficial compensator. Freezing it rather than continuing it changes validation accuracy by $+0.125$ pp ($[-0.052, 0.302]$) for the early kernel and $+0.075$ pp ($[-0.034, 0.184]$) for the late kernel. These results do not establish a final-accuracy benefit from keeping homeostasis active. They show that kernel-associated costs persist across the tested homeostatic schedules.

The final full kernels exhibit a major structural difference. Under continuing homeostasis, mean early/late receiver cosine is 0.61376 (CI $[0.61026, 0.61726]$), global relative displace-

ment is 1.28929, and mean orthogonal displacement fraction is 0.99142. Early and late DC energy fractions are 0.84387 and 0.41774; their contrast fractions are 0.15613 and 0.58226. Participation ratio rises from 1.53758 to 4.82656 and Frobenius norm from 0.79762 to 1.22726. Signed mass remains strongly positive, 0.99988 versus 0.99705: wholesale sign reversal is not the explanation. Frozen-homeostasis geometry gives nearly identical contrasts.

These are same-seed *cross-arm* comparisons, not a within-model trajectory or isolation of a rigid rotation. Direction, scale, DC/contrast balance, and spectrum change together. The findings motivate a direction intervention but do not prove that pure rotation is the unique causal variable. All four conditions have zero no-decision and all 160 seed–class mean gaps are positive. Nonetheless, negative sample margins rise from 1.508% to 3.083% for early versus late kernels with continuing homeostasis. This is a temporal decision cost within an active network, not complete inactivity concealed by a summary score.

C. Post-formation radial-only updates preserve MNIST accuracy

Table III reports the main intervention results. On MNIST, radial-only at @2 improves validation accuracy over unrestricted hidden learning by $+2.727\,\mathrm{pp}$ (CI $[2.375, 3.079]$), and test accuracy by $+2.470\,\mathrm{pp}$ ($[2.384, 2.556]$). Both sides of the temporal decision improve: $\Delta E_{\mathrm{correct}} = +0.01667$, $\Delta E_{\mathrm{wrong}} = -0.03518$, and $\Delta E_{\mathrm{gap}} = +0.05184$ (CI $[0.04705, 0.05664]$). The independent seed batch repeats the validation improvement, $+2.640\,\mathrm{pp}$ ($[2.338, 2.942]$), with a gap increase of 0.05081.

Relative to hard freeze, MNIST radial-only has a small validation difference of $+0.057\,\mathrm{pp}$ and a test difference of $-0.054\,\mathrm{pp}$, with both CIs spanning zero. The independent batch gives $+0.060\,\mathrm{pp}$ and $-0.116\,\mathrm{pp}$. These intervals are consistent with close accuracy, but a nonsignificant difference is not a proof of exact equivalence. In particular, there is a reproducible *timing* cost: radial-only correct evidence is later than hard freeze by 0.02854 in the main batch (CI $[-0.03089, -0.02620]$ for the earliness difference) and by 0.02741 in the replication. Wrong evidence is nearly unchanged, leaving a smaller gap in both batches (Fig. 4). Accuracy preservation should not be rewritten as full restoration of the hard-freeze operating point.

Partial orthogonal suppression reveals a useful range, not a strictly monotonic law. Validation accuracy for $\alpha = 1, 0.5, 0.25, 0.1, 0$ is respectively 95.077, 97.517, 97.787, 97.847, and 97.803% (five seeds). Thus reducing unrestricted directional freedom helps substantially, but the final step to zero does not improve the mean further. The sweep does not establish $\alpha = 0$ or 0.1 as a universal optimum. A proximal-reference control of strength 0.01 reaches 97.527%; this single strength is not an exhaustive comparison of regularizers.

The main radial-only arm has zero no-decision and positive gaps in all 50 seed–class combinations. Hidden/output silence in $A{-}1$ is 0.836/0.670, classifier output silence is approximately 0.86, and hidden top-decile activity share is 0.162. These sparse populations remain functional. Reference cosine stays near one, with relative drift approximately 10^{-4}. All 5,130 post-formation update steps have nonzero raw updates before projection and normalization. However, this activity and the tiny final drift do not establish meaningful post-normalization kernel adaptation.

D. N-MNIST provides a positive but bounded transfer result

On N-MNIST, unrestricted hidden learning leads to 100% no-decision by the final evaluation. This is a failure of the tested configuration, not a general property of event-based learning. Both hard freeze and radial-only prevent that collapse. Radial-only's validation difference from hard freeze is $-0.007\,\mathrm{pp}$ (CI $[-0.226, 0.211]$, seven seeds). This interval lies within the $\pm 0.5\,\mathrm{pp}$ tolerance used to assess practically similar validation accuracy. Its secondary test difference is $+0.111\,\mathrm{pp}$ ($[0.026, 0.197]$), which is small and should not be promoted into a hardware-independent superiority claim.

Unlike MNIST, radial-only improves temporal separation relative to hard freeze. Correct evidence is earlier by 0.01077 (CI $[0.00841, 0.01313]$), while wrong evidence changes by 0.00116 ($[-0.00101, 0.00333]$). The gap increases by 0.00961 ($[0.00771, 0.01151]$). All 70 seed–class mean gaps are positive. Validation no-decision is 0.131%, compared with 0.110% for hard freeze, and the negative sample-gap fractions are 1.519% and 1.540%, respectively.

This does not imply that every diagnostic improves. Hidden top-decile activity share is 0.157, but output-classifier activity concentration is higher: radial-only classifier top-decile share is 0.365 versus 0.358 for hard freeze, a paired increase of 0.00772 (CI $[0.00414, 0.01131]$). Classifier threshold quantiles are near the lower bound, not the upper bound. The unrestricted arm, in contrast, has hidden top-decile share 0.982 and complete classifier silence. Thus the successful intervention remains a concentrated temporal classifier rather than an ideally balanced population.

Mean reference cosine is 0.999993 and relative kernel displacement is 0.00308. The measured projected/raw update norm ratio averages 0.897 at the final measurement. This verifies that projection need not null the immediate update, but subsequent normalization still limits its realized effect. No linear Readout was evaluated for this comparison, so the MNIST readout/native discrepancy is not independently established here.

E. Formation stage and layer placement define the boundary

Applying the rule from initialization is not equivalent to applying it after formation. In the fresh seven-seed MNIST comparison, constraining from initialization gives 96.340% validation for radial-only and 96.167% for $\alpha = 0.1$, versus 97.814% for hard freeze after formation. Initialization-constrained $\alpha = 0.1$ is $-1.648\,\mathrm{pp}$ below that reference (CI $[-1.908, -1.388]$), despite improving over unrestricted

TABLE III

FINAL EPOCH-20 NATIVE ACCURACY AND PAIRED RADIAL-ONLY CONTRASTS. MEANS ARE PERCENTAGES; DIFFERENCES ARE PERCENTAGE POINTS WITH 95% PAIRED STUDENT-t CIS. TIMING USES NORMALIZED EARLINESS, NOT ACCURACY UNITS. MNIST REPLICATION USES INDEPENDENT SEEDS.

Dataset / batch	Condition	Seeds	Validation	Test
MNIST, main	Unrestricted	5	95.077	95.484
	Hard freeze @2	5	97.747	98.008
	Radial-only @2	5	97.803	97.954
MNIST, replication	Unrestricted	5	95.100	95.590
	Hard freeze @2	5	97.680	97.948
	Radial-only @2	5	97.740	97.832
N-MNIST	Unrestricted (100% no-decision)	7	0.000	0.000
	Hard freeze @2	7	97.024	97.173
	Radial-only @2	7	97.017	97.284

Radial minus hard freeze	Validation difference (pp)	Test difference (pp)	Timing-gap difference
MNIST, main	$+0.057\ [-0.043, 0.157]$	$-0.054\ [-0.171, 0.063]$	$-0.02763\ [-0.03151, -0.02375]$
MNIST, replication	$+0.060\ [-0.162, 0.282]$	$-0.116\ [-0.297, 0.065]$	$-0.02695\ [-0.02953, -0.02437]$
N-MNIST	$-0.007\ [-0.226, 0.211]$	$+0.111\ [0.026, 0.197]$	$+0.00961\ [0.00771, 0.01151]$

training by $+1.190\,\mathrm{pp}$ ($[0.917, 1.464]$). Radial-only remains $-1.474\,\mathrm{pp}$ below hard freeze ($[-1.584, -1.364]$).

The initialization is strongly DC-dominated: DC energy is 99.614% and participation ratio is approximately 1.008. Therefore, these results do not evaluate a well-conditioned, isotropically random dictionary of fixed directions. They establish a boundary for the tested initialization and support allowing a formation stage before restricting it.

Layer placement is equally consequential. In the independent five-seed batch, constraining only $A-1$ gives 97.740% validation, but constraining only the output cortex gives 10.130%, and constraining both gives 11.080%. Neither latter condition has any positive seed–class mean gap (0/50), despite approximately 98.5–98.7% hidden Readout test accuracy. Their output silence is only about 0.127/0.130: failure is not lack of output spikes. The same geometric constraint that stabilizes a formed hidden representation can obstruct the classifier's acquisition of class-selective timing.

Finally, a five-seed receiver-mask comparison protects half of the hidden receivers and permits radial updates in the remainder. Selecting receivers by firing or accumulated STDP utility does not establish an advantage over a random mask: their validation differences are $-0.013\,\mathrm{pp}$ (CI $[-0.168, 0.141]$) and $-0.027\,\mathrm{pp}$ ($[-0.102, 0.049]$). This negative result concerns those static selectors, not the impossibility of activity-dependent consolidation. The evaluation concerns receiver-level protection; synapse-level selection is outside its scope.

VI. DISCUSSION

A. What problem has been identified?

The evidence identifies a mismatch between continued hidden local learning and preservation of native temporal evidence. In an active MNIST network, kernel geometry can change substantially without destroying linearly readable class information. Native decisions nevertheless incur a cost that remains after kernel motion stops. The tested homeostatic controls do not eliminate that cost. This motivates *post-formation directional consolidation*, rather than the stronger assertion that all drift is harmful or that rotation alone explains every failure.

The local-learning dilemma is therefore precise: the tested hidden rule receives pre/post activity sufficient to change its representation, but no signal that certifies preservation of the downstream temporal margin. Gradient-based systems use a different information pathway, yet are not guaranteed to stop updating or to preserve past behavior either. Our experiments do not compare those learning paradigms head-to-head. They expose an engineering requirement for this local system: separate the phase that forms usable directions from the phase that must retain them while the classifier continues adapting.

B. What has the temporary solution achieved?

Radial-only provides a practical operating point that preserves native accuracy while continuing to compute projected local updates and allowing homeostatic adaptation. That is a useful reduction in reliance on *complete cortex freezing*. However, under receiver normalization it may approximate *kernel-direction freezing* extremely closely. The regulation step in Algorithm 1, the cancellation in (5), and the observed small drift make that distinction unavoidable. The present evidence does not show that radial weight changes learn new useful information. Partial suppression retains more directional freedom, but its ability to support adaptation after a distribution shift has not been tested.

The MNIST timing residual also matters. Matching accuracy while reducing the correct-class margin may leave a circuit less robust to timing noise, changed thresholds, or harder inputs. Those robustness effects are hypotheses, not measured outcomes here. Conversely, N-MNIST demonstrates that the residual is not universal: the same intervention class improves the measured timing gap there. Both observations

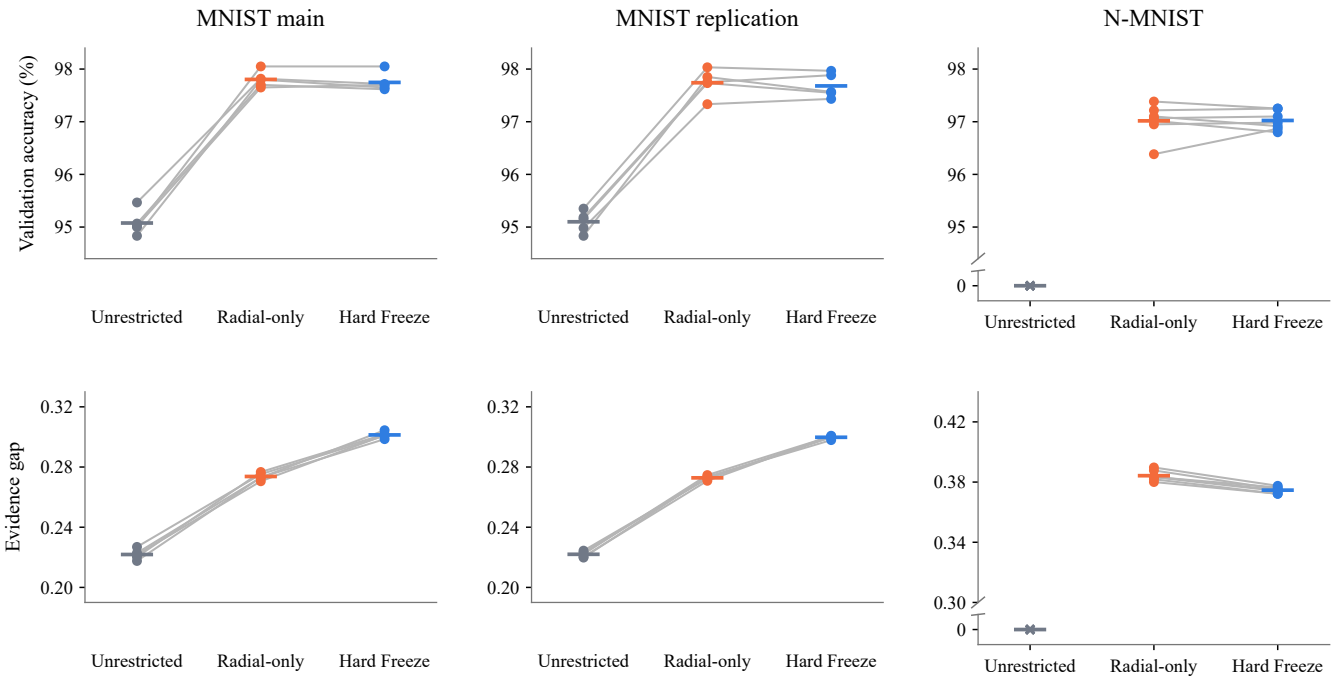


Fig. 4. Native validation accuracy and evidence gap at epoch 20. Conditions are unrestricted learning, radial-only updates, and hard freeze, from left to right. Gray lines connect matched seeds; colored points show individual runs and horizontal ticks show condition means. MNIST main and replication use the same training protocol with independent sets of five seeds; N-MNIST uses seven seeds. Top: native validation accuracy (%). Bottom: correct-minus-strongest-wrong earliness; larger gaps indicate greater temporal separation. The main axes span 4 percentage points for accuracy and 0.14 for gap; gap offsets differ between datasets. Broken N-MNIST axes retain the unrestricted zero values (crosses): all seven unrestricted runs have 100% no-decision. Their zero gap represents silence, not normal temporal decisions, and is not connected to active conditions. Hard freeze disables hidden learning and homeostasis; radial-only retains homeostasis. Seed pairing does not remove mixed-GPU confounding. Paired contrasts and CIs are given in Table III.

belong in the claim, rather than selecting the more favorable dataset as the universal explanation.

C. Limitations and next discriminating tests

The strongest unresolved mechanism question is the separation of directional change from DC/contrast and scale. The kernel/homeostasis study establishes a kernel-state-associated cost across schedules, but a shared-input cross-checkpoint temporal-drive replay and controlled direction-only versus DC-only kernel interventions are still needed for finer causal attribution. Matching GPU, software version, and replay conditions would strengthen estimates, especially for the small N-MNIST test difference.

A dedicated kernel-only-freeze versus radial-only comparison with identical homeostatic schedules is needed to determine whether projected updates add value beyond maintaining the working point. Direct logging of *post-regulation* update norms would be more informative than raw-update counters. A sequential or shifted-data task is then necessary before claiming useful continual plasticity. The present datasets are repeated stationary classification tasks; pruning, neurogenesis, and learned replacement of initially fixed directions are not evaluated methods in this paper.

The formation boundary is externally chosen, and the initialization experiment does not establish an autonomous stopping criterion. We have not measured principal-subspace trajectories or shown that any fixed epoch is universally sufficient. Results are limited to the tested two-layer architectures, encoders, and local rules. They do not establish a depth advantage, DVS-Gesture improvement, neuromorphic-hardware efficiency, or state-of-the-art accuracy. The contribution is a diagnostic account with a constrained intervention whose success and failure boundaries are visible.

VII. Conclusion

Preserving readable information is not sufficient to preserve native temporal decisions. In the studied locally plastic SNNs, hidden kernels can continue changing after coarse geometry stabilizes, and late kernel states carry a native timing cost even after their weights stop moving. Reference-direction constraints provide a simple post-formation intervention: radial-only maintains MNIST accuracy near hard freeze and prevents native N-MNIST collapse, while exposing dataset-specific timing behavior and strong stage/layer dependence. The resulting claim is deliberately bounded: limiting orthogonal hidden-kernel updates can reduce the need to freeze an entire cortex, but useful retained adaptation and rotation-specific causality remain to be demonstrated.

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